

Spatial vs. Channel Attention in Encoder-Decoder Segmentation: A Frequency-Domain Analysis of U-Net, Attention U-Net, and SEU-Net

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Abstract

Convolutional encoder-decoder networks have become the go-to approach for medical image segmentation, but evaluation almost always stops at spatial overlap metrics like Dice and Intersection over Union. These numbers tell you whether the predicted region is roughly in the right place, but they say nothing about how well a model handles fine boundaries or sharp edges. This work compares three architectures (U-Net, Attention U-Net, and SEU-Net) on the ISIC 2018 skin lesion benchmark, going beyond standard metrics by adding a frequency-domain analysis. Predicted and ground truth masks are decomposed into low, mid, and high frequency bands through radial binning in the magnitude spectrum, and per-band mean squared error is used as a complementary evaluation signal. A filtering experiment then tests robustness by passing test images through low-pass or high-pass Butterworth filters before inference, with no retraining involved. SEU-Net comes out on top in overall Dice (0.862 mean across seeds) and also achieves the lowest high-frequency spectral error (42.3 MSE), with the smallest performance drop under low-pass filtering. Attention U-Net, on the other hand, holds up best under high-pass filtering, suggesting that spatial attention gives the model a specific advantage when edge information is the dominant available signal. Overall, the results show that the two attention mechanisms do not simply differ in how much they improve performance, but in which type of frequency information they rely on to do so.

1 Introduction

Segmenting anatomical structures and lesions from medical images accurately is a core requirement for clinical decision support, treatment planning, and automated screening pipelines. Convolutional encoder-decoder networks, and U-Net in particular [1], have become the standard tool for this kind of dense prediction task. The architecture compresses input images through successive downsampling steps to build up global semantic context, then gradually restores spatial resolution through upsampling, with skip connections bringing fine-grained spatial detail back in at each decoder stage.

Despite how widely these models are used, evaluation almost always comes down to a single overlap number. The Dice coefficient measures how much of the predicted region overlaps with the ground truth relative to their combined area. Intersection over Union measures the ratio of their intersection to their union. Both tell you whether the predicted region is roughly the right shape and in the right place, but neither can distinguish between a model that produces tight, accurate boundaries and one that produces blurry or slightly shifted boundaries that happen to achieve a similar overlap score. In tasks where boundary precision is clinically important, tumour margin delineation being the obvious example, this is not a trivial distinction.

Attention mechanisms have been proposed as a way to make encoder-decoder networks more selective. Two main families have emerged. *Spatial attention*, as used in Attention U-Net [2], places a learned gating map at each skip connection that suppresses features from irrelevant spatial positions before they reach the decoder. *Channel attention*, as used in squeeze-and-excitation networks [3], instead learns a per-channel weighting that recalibrates feature maps by down-weighting uninformative filters. Both approaches improve performance on standard benchmarks, but whether they differ in terms of the spatial

frequency content they preserve has not been studied.

This work addresses that question with three contributions. First, a frequency-domain evaluation pipeline is developed that applies 2D discrete Fourier transforms to segmentation masks and computes per-band spectral error between predicted and ground truth spectra, split into low, mid, and high frequency bands via radial binning. Second, a filtering experiment tests robustness by removing either low-frequency coarse structure or high-frequency edge detail from test images before inference, with no retraining. Third, all three architectures are trained under identical conditions across two random seeds to verify that the results are stable.

The central hypothesis going in was that spatial attention should produce lower high-frequency spectral error than channel attention, since spatial gating directly controls which parts of the skip features (the same features that carry fine boundary information from encoder to decoder) get passed through. The experimental results tell a more nuanced story: the hypothesis holds in the filtering domain but not in the spectral error domain, and understanding why that split occurs is one of the more interesting outcomes of the work.

2 Related Work

2.1 U-Net and Encoder–Decoder Variants

U-Net [1] introduced the encoder–decoder with symmetric skip connections as a practical solution to the problem of dense prediction from limited annotated data. The contracting path builds up hierarchical feature representations through repeated convolution and pooling. The expansive path then restores spatial resolution step by step, pulling in the corresponding encoder feature map at each level through a skip connection to recover fine spatial detail that would otherwise be lost during downsampling.

A number of extensions have built on this foundation. U-Net++ [4] replaced the direct skip connections with nested dense connections and intermediate convolutional blocks, gradually bridging the semantic gap between encoder and decoder features rather than jumping across it in one step. MultiResUNet [5] swapped out the standard double convolution blocks for multi-scale residual blocks that process features at several spatial scales simultaneously within each encoder stage. TransUNet [6] added transformer self-attention at the bottleneck to capture long-range dependencies that local convolutions simply cannot reach. All of these variants keep the core encoder–decoder structure and skip connections intact while changing how features are extracted or combined along the way.

2.2 Attention Mechanisms in Segmentation

Attention was originally developed for sequence-to-sequence models, where it allowed decoders to selectively focus on the most relevant parts of the encoded input at each step [7]. Bringing that idea into image segmentation has taken two main forms.

Attention U-Net [2] added attention gates directly at each skip connection. Each gate receives a gating signal from the decoder side, which carries high-level information about what the network is currently trying to reconstruct, and the encoder feature map, which carries fine spatial detail. From these two inputs it produces a spatial attention coefficient that gets multiplied onto the skip features before they are concatenated into the decoder. In practice this means the model learns to focus on relevant spatial regions and suppress background noise at each decoder stage. The approach was originally validated on abdominal organ segmentation and showed consistent gains over the plain U-Net baseline.

Squeeze-and-excitation blocks [3] take a different angle, applying attention in the channel dimension rather than spatially. Global average pooling collapses the spatial dimensions of a feature map down to a single value per channel. A small two-layer fully connected network with a sigmoid output then produces per-channel recalibration weights, which are multiplied back onto the original feature map. The result is that the network learns which feature channels are most informative for the task and turns up or down their relative contribution. SE blocks have been added to U-Net encoders for skin lesion segmentation [10] and a range of other medical imaging tasks, generally giving modest but consistent improvements.

2.3 Frequency-Domain Methods in Vision

Fourier analysis has been used to probe how convolutional networks handle spatial frequency information. Geirhos et al. [8] showed that CNNs trained on ImageNet develop a strong texture bias, leaning on high-frequency pattern matching more than on low-frequency structural cues. Park and Kim [9] found that vision transformers tend to capture lower-frequency features more effectively than convolutional networks, which are pulled toward local high-frequency patterns by the nature of their small receptive fields.

In medical image segmentation, frequency-domain ideas have mostly been used on the input side, for example incorporating Fourier-based operations into network architectures to improve generalisation across imaging domains. Using Fourier decomposition on the output side, to evaluate how well different model predictions reconstruct boundary-level versus shape-level information, has received much less attention. That is the role it plays in this work.

3 Methodology

3.1 Architecture Descriptions

All three architectures share an identical backbone: four encoder blocks, a bottleneck, and four decoder blocks. Each encoder block runs the input through two consecutive convolution–batch normalisation–ReLU layers and then applies 2×2 max pooling to halve the spatial resolution. Each decoder block uses transposed convolution to upsample, concatenates the corresponding skip connection, and applies another double convolution. A final 1×1 convolution produces the single-channel output. The channel depth follows 64, 128, 256, 512 through the encoder, reaching 1024 at the bottleneck.

U-Net. The baseline follows the standard design with batch normalisation after each convolution. Skip connections pass encoder feature maps directly to decoder inputs with no gating or recalibration of any kind. This serves as the no-attention reference point and has 31.0 million trainable parameters.

Attention U-Net. An attention gate is placed at each of the four skip connections. Given a gating signal $\mathbf{g}$ from the decoder and a skip feature map $\mathbf{x}$ from the encoder, the gate computes:

$$\alpha = \sigma(W_\psi \cdot \text{ReLU}(W_g \mathbf{g} + W_x \mathbf{x})) \quad (1)$$

where σ is the sigmoid function and W_g, W_x, W_ψ are learned 1×1 convolutions. The result $\alpha \in (0, 1)^{H \times W}$ is a spatial attention map that gets multiplied element-wise onto $\mathbf{x}$ before concatenation. The idea is that the decoder’s high-level signal guides the gate to suppress encoder features from spatial regions that are not relevant to the current prediction step. This model has 31.4 million parameters.

SEU-Net. A squeeze-and-excitation block is added after each encoder double convolution, including at the bottleneck. For a feature map $\mathbf{F} \in \mathbb{R}^{C \times H \times W}$, the SE block computes:

$$\mathbf{s} = \sigma(W_2 \cdot \text{ReLU}(W_1 \cdot \text{GAP}(\mathbf{F}))) \quad (2)$$

where GAP is global average pooling across spatial dimensions and W_1, W_2 form a bottleneck excitation with reduction ratio 16. The vector $\mathbf{s} \in \mathbb{R}^C$ is then multiplied channel-wise back onto $\mathbf{F}$. The decoder is identical to the plain U-Net decoder. This model has 31.2 million parameters.

The core difference between the two attention variants comes down to the axis they operate on: Attention U-Net asks *where* to look by gating spatially at skip connections, while SEU-Net asks *what* to look at by recalibrating feature channels in the encoder.

3.2 Frequency Analysis Framework

For each test image the 2D discrete Fourier transform is applied to both the ground truth binary mask M_{GT} and the predicted binary mask M_{pred} :

$$\hat{M}(u, v) = \sum_{x=0}^{H-1} \sum_{y=0}^{W-1} M(x, y) e^{-2\pi i (ux/H + vy/W)} \quad (3)$$

The magnitude spectrum $|\hat{M}|$ is shifted so that the DC component sits at the center of the image. Frequency space is then divided into three concentric rings based on Euclidean distance r from DC. For 256×256 images the maximum radius is $r_{\max} = \sqrt{128^2 + 128^2}$, and the bands are spaced evenly:

- **Low band** ($r < r_{\max}/3$): coarse lesion shape and overall location.
- **Mid band** ($r_{\max}/3 \leq r < 2r_{\max}/3$): intermediate texture and moderate-scale structure.
- **High band** ($r \geq 2r_{\max}/3$): fine edges, sharp boundaries, and small-scale detail.

Within each ring, per-band mean squared error between the predicted and ground truth magnitude spectra is computed:

$$\text{MSE}_b = \frac{1}{|B_b|} \sum_{(u,v) \in B_b} \left(|\hat{M}_{GT}(u,v)| - |\hat{M}_{pred}(u,v)| \right)^2 \quad (4)$$

where B_b is the set of frequency coordinates belonging to band b . This is averaged over all 390 test images to give a single per-band error value per model.

3.3 Filtering Experiment Design

To test how each model handles frequency-degraded inputs, Butterworth filters of order $n = 2$ are applied to test images before inference. No models are retrained on filtered data. The low-pass transfer function is:

$$H_{LP}(r) = \frac{1}{1 + (r/f_c)^{2n}} \quad (5)$$

and the high-pass version is simply $H_{HP}(r) = 1 - H_{LP}(r)$. The cutoff f_c is expressed as a fraction of $r_{\max}$. For high-pass filtered images, the per-channel mean brightness is added back after reconstruction; without this step the DC removal leaves the image nearly black, which is not a useful input for inference. Four conditions are tested:

- **LP-0.05**: aggressive low-pass blur, only coarse shape survives.
- **LP-0.10**: moderate blur, more detail preserved.
- **HP-0.02**: gentle high-pass, lesion boundary still visible.
- **HP-0.04**: stronger high-pass, coarse lesion shape largely gone.

Cutoffs were chosen by looking at the filtered images directly and confirming that some lesion signal was still visible in each case before running inference.

3.4 Training Protocol

All three models are trained with the same hyperparameters throughout. The loss function combines binary cross-entropy with soft Dice loss:

$$\mathcal{L} = \mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{Dice}} \quad (6)$$

Optimisation uses Adam [12] with a fixed learning rate of 10^{-4} and no schedule. Batch size is 64 and training runs for 30 epochs. The checkpoint with the best validation Dice is saved and used for all subsequent evaluation. Each architecture is trained twice using seeds 42 and 123, with randomness fixed across Python, NumPy, PyTorch, and CUDA to make the runs reproducible. All training was done on an NVIDIA A100-SXM4-40GB GPU.

4 Experiments

4.1 Dataset

Experiments use the ISIC 2018 Task 1 skin lesion segmentation benchmark [11]. The dataset contains 2,594 dermoscopy images with binary segmentation masks annotated by clinical experts, marking the boundary of the primary skin lesion. Images vary substantially in resolution, illumination, lesion size, and morphology, ranging from small symmetric nevi to large irregular melanomas with diffuse borders.

4.2 Data Splits

The official ISIC 2018 test set does not include released ground truth masks. A 70/15/15 split is applied to the 2,594 labelled images, producing 1,815 training, 389 validation, and 390 test images. Split indices are saved to a JSON file before any training and shared across all six model runs. All evaluation is on the same 390 test images.

4.3 Preprocessing and Augmentation

Images are resized to 256×256 using bilinear interpolation; masks use nearest-neighbour and are thresholded at 0.5. Pixel values are normalised to $[0, 1]$. Training augmentations are random horizontal flip, vertical flip, and 90/180/270-degree rotation (each with probability 0.5), applied identically to image and mask. No augmentation is applied at test time.

4.4 Evaluation Metrics

Standard metrics computed on binary predictions thresholded at 0.5: Dice, IoU, Precision, Recall. Per-band spectral MSE as in Eq. 4 for frequency analysis. Dice and Dice drop for the filtering experiment.

5 Results

5.1 Spatial Metrics

Table 1 reports Dice, IoU, Precision, and Recall for all three architectures on the 390-image test set for both seeds, and the mean across seeds. All models achieve Dice above 0.855. The ranking is stable across both seeds: SEU-Net > Attention U-Net > U-Net. Mean Dice values are 0.858, 0.860, and 0.862 respectively.

Table 1: Test-set spatial metrics. Values are mean $\pm$ std over the 390 test images. The *mean* row averages the two seeds per architecture.

Model	Seed	Dice	IoU	Precision	Recall
U-Net	42	0.855 $\pm$ 0.159	0.773 $\pm$ 0.188	0.871 $\pm$ 0.160	0.893 $\pm$ 0.172
	123	0.860 $\pm$ 0.161	0.781 $\pm$ 0.191	0.881 $\pm$ 0.168	0.885 $\pm$ 0.167
	<i>mean</i>	<i>0.858</i>	<i>0.777</i>	<i>0.876</i>	<i>0.889</i>
Attn U-Net	42	0.865 $\pm$ 0.161	0.788 $\pm$ 0.189	0.892 $\pm$ 0.170	0.884 $\pm$ 0.162
	123	0.855 $\pm$ 0.167	0.775 $\pm$ 0.193	0.886 $\pm$ 0.151	0.881 $\pm$ 0.188
	<i>mean</i>	<i>0.860</i>	<i>0.782</i>	<i>0.889</i>	<i>0.883</i>
SEU-Net	42	0.866 $\pm$ 0.170	0.792 $\pm$ 0.191	0.887 $\pm$ 0.157	0.889 $\pm$ 0.188
	123	0.858 $\pm$ 0.175	0.781 $\pm$ 0.197	0.885 $\pm$ 0.139	0.887 $\pm$ 0.199
	<i>mean</i>	<i>0.862</i>	<i>0.787</i>	<i>0.886</i>	<i>0.888</i>

Figure 1 visualises Dice, IoU, Precision, and Recall. Differences between models are under 0.5% in mean Dice, which is expected given the identical backbone and training setup.

Spatial Metrics Comparison — Mean across seeds

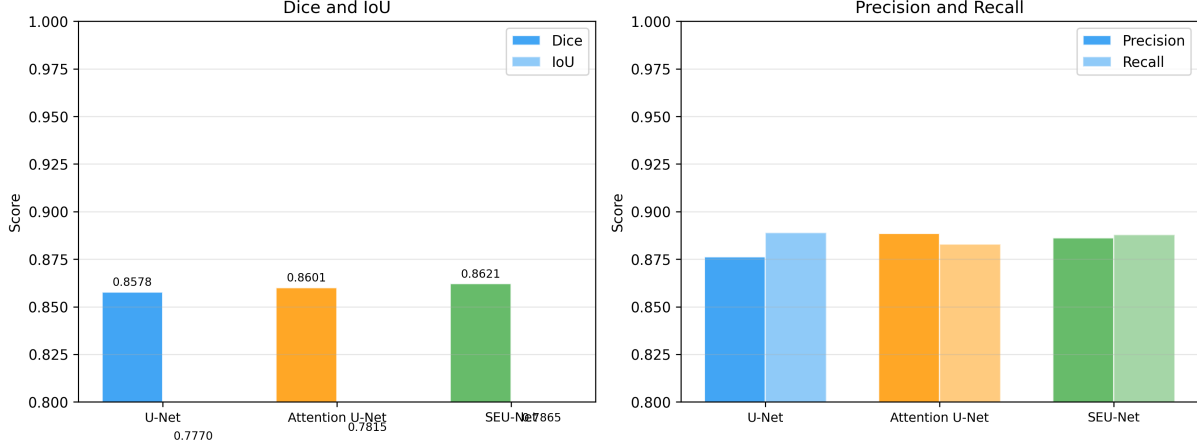


Figure 1: Spatial metrics averaged across seeds. Left: Dice and IoU. Right: Precision and Recall. Blue = U-Net, orange = Attention U-Net, green = SEU-Net.

Figure 2 shows training curves for all six runs. The two seeds for each architecture track closely, confirming that the ranking is not due to initialisation differences.

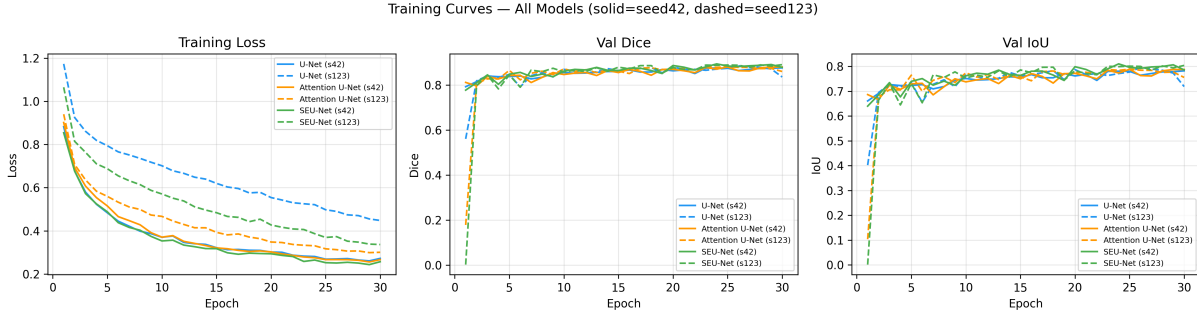


Figure 2: Training loss, validation Dice, and validation IoU over 30 epochs for all models and seeds. Solid = seed 42, dashed = seed 123. Colors as in Figure 1.

5.2 Frequency Domain Analysis

Table 2 reports per-band spectral MSE averaged over the 390 test images and across both seeds. The high-frequency column is the most interpretable, measuring boundary-level reconstruction fidelity.

Table 2: Per-band spectral MSE between predicted and ground truth magnitude spectra, averaged across seeds. Lower is better.

Model	Low MSE	Mid MSE	High MSE
U-Net	8422.1	78.0	45.3
Attn U-Net	8526.2	75.8	43.6
SEU-Net	7994.2	74.1	42.3

The high-frequency MSE ranking mirrors the spatial Dice ranking: SEU-Net achieves the lowest error (42.3), Attention U-Net is in the middle (43.6), and U-Net has the highest (45.3). In the low-frequency band, Attention U-Net has a slightly higher MSE than U-Net despite better Dice, indicating some difference in how the two models reconstruct coarse shape.

Figure 3 shows the radial frequency error profiles. The left panel shows grouped bars per band averaged across seeds. The right panel shows individual seed lines confirming consistency.

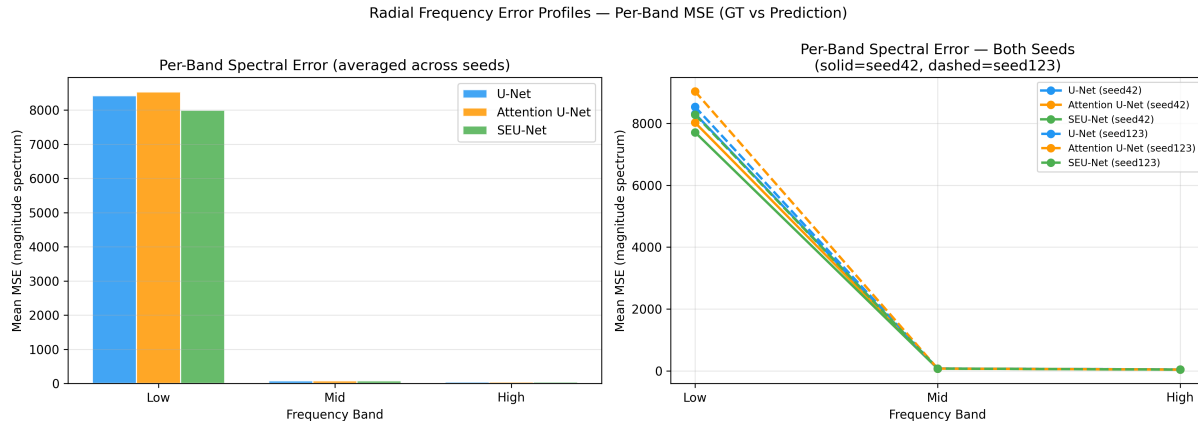


Figure 3: Radial frequency error profiles. Left: per-band MSE averaged across seeds. Right: per-seed lines (solid = seed 42, dashed = seed 123). The high-frequency band is the primary indicator of boundary reconstruction fidelity.

5.3 Filtering Experiment

Table 3 reports Dice on filtered inputs and the drop from the unfiltered seed-42 baseline for all four conditions.

Table 3: Dice under frequency filtering. Baseline from seed 42. Drop = baseline Dice minus filtered Dice; positive = worse performance. LP = low-pass, HP = high-pass, f_c as fraction of $r_{\max}$.

Condition	U-Net		Attn U-Net		SEU-Net	
	Dice	Drop	Dice	Drop	Dice	Drop
Baseline (unfiltered)	0.855	—	0.865	—	0.866	—
LP $f_c = 0.05$	0.819	+0.036	0.838	+0.027	0.836	+0.030
LP $f_c = 0.10$	0.830	+0.026	0.843	+0.021	0.849	+0.017
HP $f_c = 0.02$	0.669	+0.186	0.725	+0.140	0.664	+0.203
HP $f_c = 0.04$	0.360	+0.496	0.418	+0.447	0.390	+0.477

Under low-pass filtering U-Net degrades most at both cutoffs. At LP-0.10, U-Net drops 0.026 while SEU-Net drops only 0.017, roughly a one-third reduction in degradation. Under high-pass filtering the ranking reverses: Attention U-Net is clearly the most robust, losing 0.140 Dice at HP-0.02 compared to 0.186 for U-Net and 0.203 for SEU-Net.

Figure 4 visualises these results. The right panel showing Dice drops is particularly informative: the models show clearly different patterns for low-pass versus high-pass conditions.

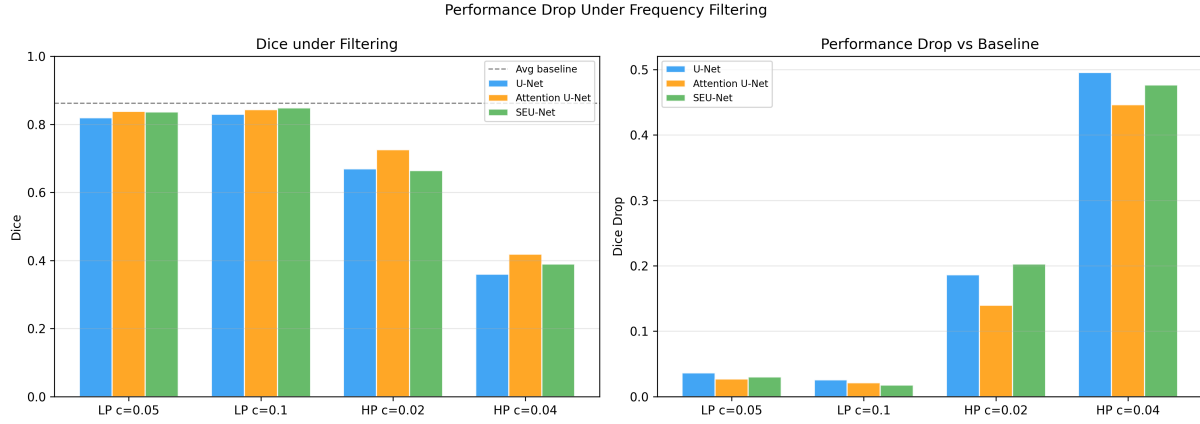


Figure 4: Filtering experiment results. Left: absolute Dice under each condition. Right: Dice drop relative to the unfiltered baseline. LP = low-pass, HP = high-pass.

5.4 Qualitative Analysis

Figure 5 shows boundary overlays for four test images where U-Net and SEU-Net differ most in Dice. White contours are ground truth; coloured contours are predictions. The selection method picks the most extreme cases of disagreement, which includes instances where one model fails entirely and produces no visible prediction. On cases where all models segment successfully, SEU-Net (green) tends to track the ground truth boundary more closely on irregular lesion shapes, while Attention U-Net (orange) produces smoother contours.

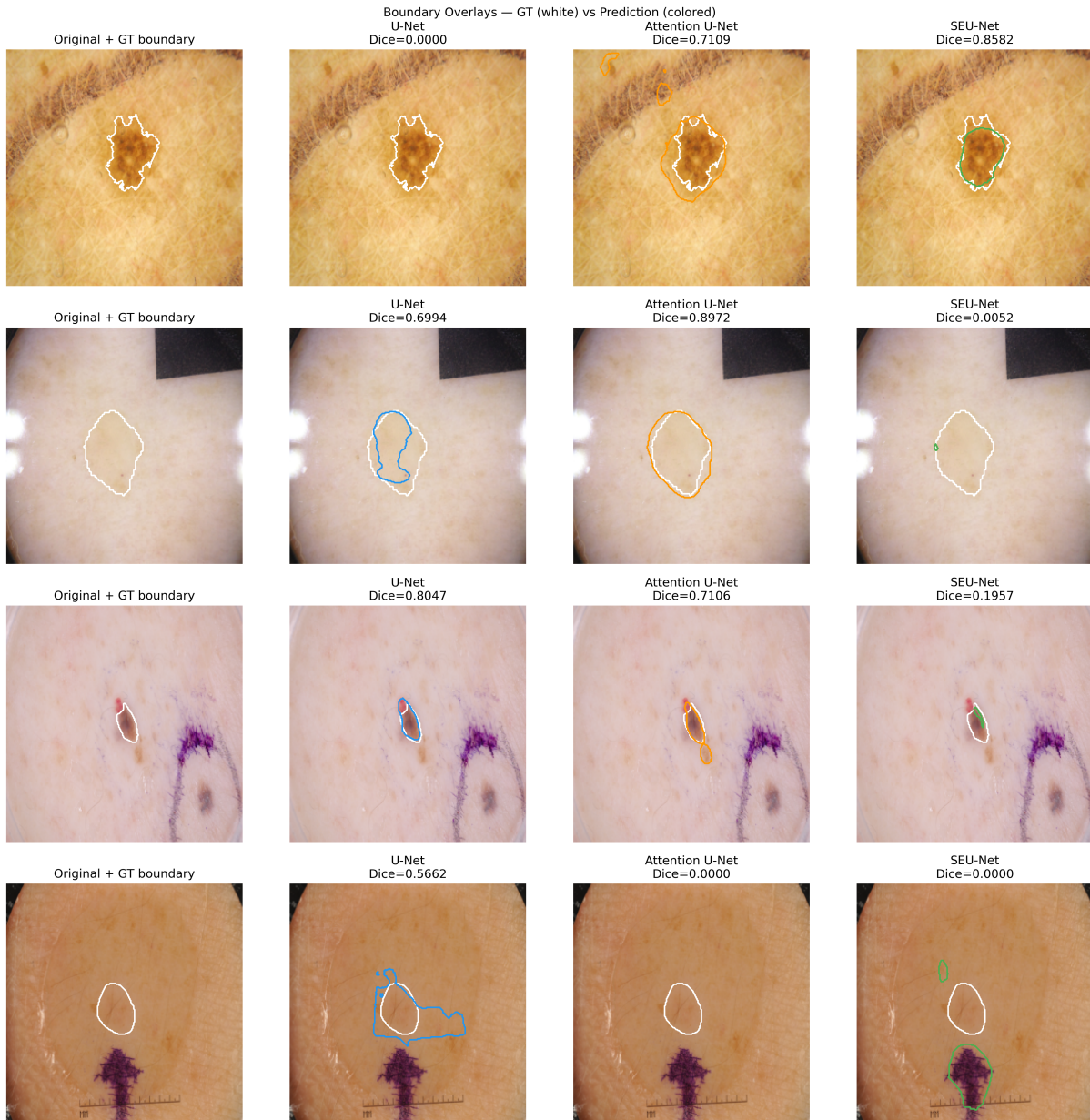


Figure 5: Boundary overlays for four selected test images with the largest Dice difference between U-Net and SEU-Net. White = ground truth. Blue = U-Net. Orange = Attention U-Net. Green = SEU-Net.

Figure 6 shows frequency error heatmaps for the same images. Brighter regions indicate larger spectral discrepancies between predicted and ground truth spectra.

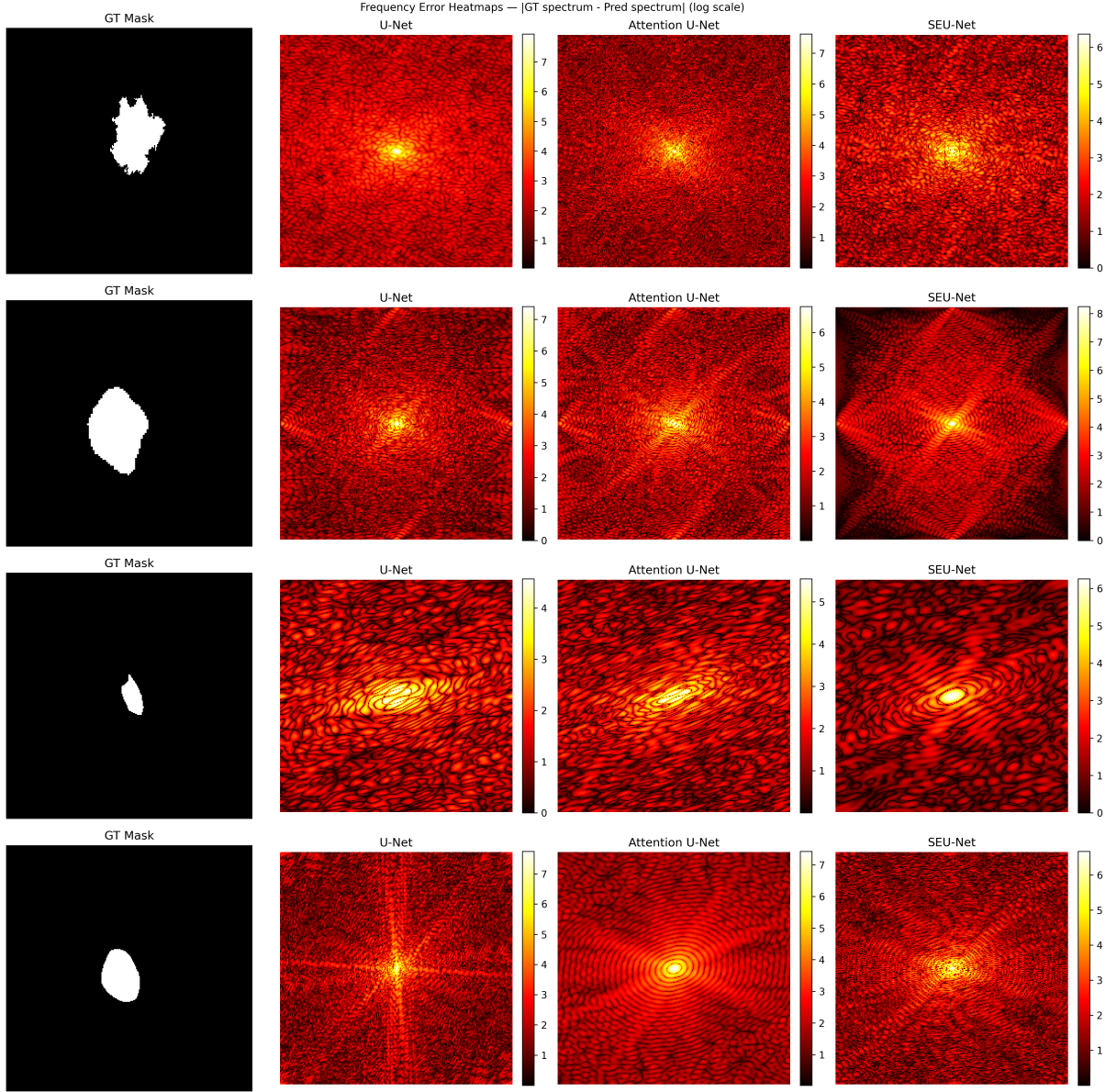


Figure 6: Frequency error heatmaps for the same four images. Pixel intensity is $\log(1 + |\hat{M}_{GT} - \hat{M}_{pred}|)$. Brighter areas indicate larger spectral error. The high-frequency periphery of each heatmap corresponds to boundary-level error.

6 Discussion

6.1 Overall Performance and Stability

All three models land within a narrow Dice range of 0.858 to 0.862, consistent with reported results for U-Net scale models on ISIC 2018 trained from scratch. The ranking (SEU-Net above Attention U-Net above plain U-Net) holds across both seeds and appears consistently in the spatial metrics, the frequency analysis, and the filtering experiment. With only two runs per architecture formal statistical testing is not possible, but the directional consistency across every evaluation dimension gives reasonable confidence that the differences reflect genuine architectural properties.

The practical significance of the absolute numbers is limited. A Dice gap of under 0.005 would not matter in a clinical setting. The contribution here is not a claim that one architecture is strictly better, but a

clearer picture of how the three models differ along dimensions that the aggregate overlap metric cannot show.

6.2 Frequency Domain Results

The starting hypothesis was that spatial attention should produce lower high-frequency spectral error than channel attention, since spatial gating controls what passes through the skip connections, the main route for fine boundary information from encoder to decoder. The spectral MSE results do not support this. SEU-Net achieves the lowest high-frequency error without any spatial gating, and Attention U-Net sits in the middle.

One explanation is that SE blocks recalibrate channels globally at every encoder stage, suppressing uninformative filter responses regardless of spatial location. This produces cleaner feature maps throughout the hierarchy that carry through to the predictions. Attention gates only operate at skip connections and can only suppress by spatial position, not by channel content. Noise that overlaps spatially with the lesion boundary passes through alongside the useful signal.

The slightly higher low-frequency MSE for Attention U-Net compared to plain U-Net is a smaller but plausible side effect: attention gates may occasionally suppress coarse shape information in the skip connection, slightly degrading low-frequency reconstruction even as overall Dice improves.

6.3 Filtering Results and Partial Support for the Hypothesis

The filtering experiment partially rehabilitates the original hypothesis. Under high-pass filtering, which removes the coarse lesion blob and leaves only edges, Attention U-Net holds up best. At HP-0.02 it loses 0.140 Dice, while U-Net loses 0.186 and SEU-Net loses 0.203. Spatial attention gates, trained to focus on lesion boundaries during normal training, transfer that selectivity to edge-only inputs where coarse colour and shape cues are absent.

SEU-Net degrades more than plain U-Net under high-pass conditions. The SE excitation weights are tuned on normal images where the coarse colour blob is a dominant signal. When that signal disappears, the channel weighting strategy becomes less effective rather than more.

The low-pass results fit the spectral MSE findings. SEU-Net is most robust to the removal of boundary detail, matching its lower high-frequency spectral error on unfiltered images. It can identify the lesion from coarse structure alone.

The two sets of results point to a consistent picture: spatial and channel attention rely on complementary types of frequency information. SE blocks improve feature selectivity globally, helping under normal conditions. Attention gates learn spatial selectivity that proves valuable specifically when edge information is the dominant signal. The two properties suggest a combined architecture could offer the benefits of both.

7 Conclusion

This work compared three encoder-decoder segmentation architectures through a frequency-domain lens on ISIC 2018, combining spatial metrics with per-band spectral error and a Butterworth filtering experiment that selectively removed high-frequency or low-frequency content from test images before inference.

SEU-Net achieves the best overall Dice (0.862, mean across seeds), the lowest high-frequency spectral error (42.3 MSE), and the smallest degradation under low-pass filtering, pointing to channel attention improving global feature discrimination. Attention U-Net holds up best under high-pass filtering, partially supporting the original hypothesis: spatial attention provides a real advantage when edge information is the primary available signal, not under normal conditions. Plain U-Net is the most sensitive to low-pass degradation across the board.

The broader point is that spatial and channel attention differ in which part of the frequency spectrum they rely on, and this matters for architecture selection in tasks where specific types of image information may be degraded or absent.

Limitations

Each model was trained with two seeds, enough to check directional consistency but not for formal statistical testing. The filtering baseline uses seed 42 only. The frequency analysis discards phase information, and the three-band decomposition is arbitrary, and different divisions could shift the relative results.

Future Work

The most direct follow-up is a combined architecture with SE blocks in the encoder and attention gates at skip connections. The frequency pipeline could be applied to other benchmarks such as polyp or retinal vessel segmentation to test generalisability. Extending the filtering experiment to noise and compression artefacts would connect the findings to real-world imaging conditions.

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